

SYLLABUS: COSC 594/650 – FALL 2026

Course Name: Optimization for Machine Learning

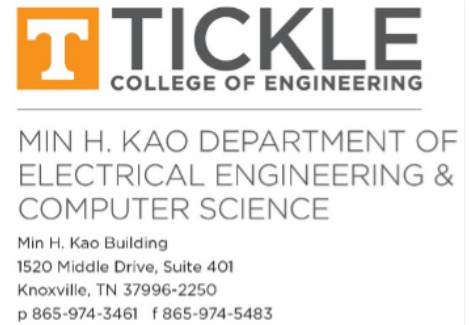
Lecture Time: Mon, Wed, Fri, 9:10 am - 10:00 am

Lecture Location: Perkins Hall 108

Instructor: Dongsheng Ding (dongshed@utk.edu)

Office: Min Kao Building 315

Office Hours: Fri, 11:30 am – 1:00 pm (or by appointment)



Course Description: This course introduces the optimization foundations for modern machine learning (or machine decision-making/artificial intelligence). Topics include first-order, stochastic, constrained, non-smooth, minimax, and policy optimization, with applications to generative modeling, and reinforcement learning. The course emphasizes how operational, safety, robustness, and alignment requirements are translated into optimization problems. Students will read, reproduce, and critique recent optimization research for machine learning, culminating in a conference-format research paper and reproducible software artifact.

Learning Outcomes: By the end, students should be able to:

- Formulate machine decision-making problems using mathematical optimization language.
- Derive optimality conditions and use convexity, smoothness, duality, and constraint qualifications correctly.
- Analyze convergence and complexity of representative optimization algorithms.
- Implement and diagnose optimization algorithms using objective, stationarity, and constraint-violation metrics.
- Read research papers critically, reproduce central results, and identify defensible research gaps.
- Develop theoretical or empirical research that validates falsifiable claims.
- Produce conference-format manuscripts, reproducible repositories, peer reviews, and rebuttal.
- Use AI tools transparently while independently verifying mathematical claims, code, experiments, and citations.

The final outcome will be a submission-ready research package, while submission will depend on whether the project meets a sufficient novelty and evidence threshold.

Teaching Assistant:

- Anik Saha (asaha15@vols.utk.edu)

Office and office hours: Min Kao Building 349, Thur, 12:00 pm - 1:00 pm

Prerequisites: Students are expected to have a foundation in linear algebra, probability, and calculus, as well as basic knowledge of Python. Measure-theoretic probability is not required.

Textbooks: There is no required textbook. The following are recommended references:

- Boyd S, Vandenberghe L. Convex optimization. Cambridge university press; 2004. A free electronic version is available at https://web.stanford.edu/~boyd/cvxbook/bv_cvxbook.pdf
- Bubeck S. Convex Optimization:: Algorithms and Complexity. Foundations and Trends in Machine Learning; 2015. A free electronic version is available at <https://arxiv.org/abs/1405.4980>
- Shalev-Shwartz S, Ben-David S. Understanding machine learning: From theory to algorithms. Cambridge university press; 2014. A free electronic version for personal use is available at <https://www.cs.huji.ac.il/~shais/UnderstandingMachineLearning/copy.html>

Course Website: CANVAS in your myutk account.

Tentative Course Outline:

- Week 1: Aug 17, Aug 19, Aug 21
 - Course overview and objectives
 - Preliminaries: Calculus, linear algebra, and probability basics
 - Unconstrained optimization: Necessary optimality conditions, smooth vs nonsmooth optimization
- Week 2: Aug 24, Aug 26, Aug 28
 - Sufficient optimality conditions, convex vs nonconvex optimization
 - Gradient descent method, convergence analysis, and oracle complexity analysis
- Week 3: Aug 31, Sep 2, Sep 4
 - Momentum and accelerated methods
 - Convergence analysis
- Week 4: Sep 9, Sep 11
 - Constrained optimization: Necessary optimality conditions, Lagrangian duality
 - Dual method and convergence analysis
- Week 5: Sep 14, Sep 16, Sep 18 (no class on Sep 14 due to conference)
 - Primal-dual method and convergence analysis
 - Augmented Lagrangian method and convergence analysis
- Week 6: Sep 21, Sep 23, Sep 25
 - Minimax optimization: Solution concepts, necessary optimality conditions
 - Gradient-based methods and convergence analysis
- Week 7: Sep 28, Sep 30, Oct 2
 - Stochastic optimization: Stochastic gradient descent method
 - Mini-batching, variance reduction, adaptive methods
 - In-class midterm
- Week 8: Oct 7, Oct 9

- Nonsmooth optimization: Proximal, projected, and mirror methods
- Week 9: Oct 12, Oct 14, Oct 16
 - Nonconvex optimization: Stationarity, PL conditions, and escaping saddle points
- Week 10: Oct 19, Oct 21, Oct 23
 - Neural network optimization
- Week 11: Oct 26, Oct 28, Oct 30
 - Optimization for diffusion/flow models
- Week 12: Nov 2, Nov 4, Nov 6 (no classes this week due to conference)
- Week 13: Nov 9, Nov 11, Nov 13
 - Optimization for large language models
- Week 14: Nov 16, Nov 18, Nov 20
 - Policy optimization in reinforcement learning
- Week 15: Nov 23
 - Student presentations
- Week 16: Nov 30
 - Student presentations

Grading: 5% Class participation, 10% Lecture scribing, 20% Homework (5 assignments, each 4%), 20% In-class midterm, 45% Project (10% proposal and research question, 5% technical checkpoint, 10% presentation, 5% peer review and rebuttal, 15% final report).

- Class participation includes asking questions, engaging in discussions, and providing feedback.
- Students are required to scribe notes for an assigned lecture or optimization topic. This assignment is intended to develop technical writing and verification skills. The notes should be sufficiently clear and self-contained for a student who missed the lecture to understand the material. Each set of scribed notes is due one week after the assigned lecture, and will be posted on CANVAS.
- There are five homework assignments that are equally weighted. Discussions with others are allowed, but all answers must be completed independently. Homework is due one week after it is assigned. **Late submission will not be accepted.**
- Project aims to research an optimization problem for machine decision-making. You are required to identify a machine decision-making problem and formulate it as a mathematical optimization problem. You are encouraged to talk to your advisors about a potential problem. If you don't have advisors, you are welcome to talk to us, and we are happy to make suggestions and help you refine your initial ideas. Three key milestones are: (1) Proposal; (2) Presentation; (3) Report. Between them, there are two check points: (1) Technical checkpoint; and (2) Peer review and rebuttal. See expectations for each step in Project Guideline. The final report is due on the last day of this semester. **Late submission will not be accepted.**

- The project should be growable into a publishable paper at a top-tier conference or journal. Students in a higher-numbered section are expected to make a stronger original theoretical or empirical contribution, while students in the lower-numbered section could complete a meaningful and rigorous extension.

Use of AI: This course treats AI tools (e.g., ChatGPT, Claude, Gemini, GitHub Copilot, and similar systems) potentially useful research assistants, but not as substitutes for mathematical reasoning, scientific judgment, or individual learning. Unless otherwise stated for a particular assignment, students may use AI tools subject to the following requirements.

- **Responsibility and understanding.** You are responsible for every statement, derivation, citation, line of code, experimental result, and conclusion that you submit. You must be able to reproduce, explain, and defend your work without assistance from an AI tool. AI-generated output is not evidence that a claim is correct.
- **Disclosure.** Every submission involving AI assistance must include an **AI Use Statement** identifying: (1) the tool and model used; (2) the date or dates of use; (3) the purpose for which it was used; (4) the portions of the submission affected; and (5) how the output was checked, corrected, or revised. Representative prompts and any materially used outputs should be included in an appendix. Students should retain their complete AI interaction records until the assignment has been graded. If no AI assistance was used, write: “No AI tools were used in preparing this submission.” Ordinary, non-generative spell-checking does not need to be disclosed.
- **Permitted assistance.** Appropriate uses may include brainstorming, requesting explanations, obtaining feedback on an independently developed solution, improving organization or grammar, generating initial code, debugging, constructing test cases, checking algebra, and identifying potentially relevant literature. Assignment-specific instructions may restrict these uses.
- **Prohibited uses.** You may not submit AI-generated material that you have not independently understood and verified. You may not use AI to fabricate citations, proofs, data, experimental results, or claims of novelty. Directly copying AI-generated text, mathematics, or code without appropriate acknowledgment, verification, and substantive student contribution is prohibited.
- **Mathematics and citations.** All AI-assisted derivations must be checked independently, with assumptions and intermediate steps made explicit. References suggested by an AI tool must be located, read, and verified from the original source before being cited. An AI-generated answer is not a substitute for citing the relevant paper, book, software documentation, or dataset.
- **Lecture scribing.** AI tools may be used to organize, edit, or check scribed notes. The student remains responsible for ensuring that the notes accurately represent the lecture, contain complete derivations, and are understandable and self-contained. Course recordings, unpublished slides, or another student’s contributions may not be uploaded to an external AI service without permission.
- **Homework.** AI tools may provide hints, critique a proposed argument, check a completed solution, or help develop and debug initial code. The final mathematical formulation, reasoning, derivation, and explanation must reflect the student’s own understanding. Submitting an AI response as a solution, even with minor rewriting, is prohibited.
- **Course project.** AI tools may be used throughout the research process, including for brainstorming, coding, editing, and diagnostic feedback. However, the research question, claimed contribution, technical decisions, validation methodology, and interpretation of results must be developed and defended by the student. The final manuscript and repository must contain an **AI Use Statement**. AI-assisted code, text, or figures must be verified, and applicable attribution, licensing, data-use, and conference or journal policies must be followed.

- **Examinations.** AI tools are prohibited during the in-class midterm and any other examination or closed assessment unless their use is explicitly authorized in writing.
- **Privacy and research confidentiality.** Do not enter confidential research, personally identifiable information, unpublished work belonging to others, restricted datasets, assessment materials, or copyrighted course content into an external AI system without authorization. Students working with an advisor or research group should obtain permission before uploading unpublished project materials.
- **Verification of mastery.** The instructor or teaching assistant may ask any student or project team to explain a submitted derivation, reproduce part of a solution, demonstrate code, or answer questions about an AI-assisted submission. An inability to demonstrate the knowledge represented in the submission may result in reassessment under the relevant grading rubric. Suspected undisclosed or unauthorized AI use will be addressed through the University's academic misconduct procedures.
- **Assignment-specific rules.** Instructions for an individual assignment or examination take precedence over this general policy. No student will be required to purchase access to a paid AI service. If you are uncertain whether a particular use is permitted, ask the instructor or teaching assistant before submitting the work.

ADA and Accessibility: If you need any accommodation, please let me know as soon as possible. Please find the university services at <https://sds.utk.edu>

Honor Statement: All students are subject to the Student Conduct & Community Standards: <https://studentlife.utk.edu/studentconduct/academic-dishonesty/>